



Question 1) Unitary operators are those for which $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{1}$.

- Prove that if $\hat{U}$ is unitary, that its eigenvalues have absolute value 1.
- Writing $\hat{U} = e^{i\hat{H}}$ for some operator $\hat{H}$, show that $\hat{U}$ is unitary when $\hat{H}$ is Hermitian. (Expand $e^{i\hat{H}}$).
- For any two orthonormal bases $\{|a_i\rangle\}$ and $\{|b_i\rangle\}$, show that there exists a unitary operator $\hat{U}$ such that $\hat{U}|a_i\rangle = |b_i\rangle$ for all i . What is it?
- Show that as far as all the matrix elements of an operator are concerned, a **unitary** transformation of bases $|a_i\rangle \mapsto |b_i\rangle = \hat{U}|a_i\rangle$ is equivalent to a **similarity transformation** of the operator, i.e., if $\hat{A} \mapsto \hat{A}' = \hat{U}\hat{A}\hat{U}^\dagger$.
- Show that a unitary transformation of bases preserves:
 - inner products, i.e., for any vectors $|\psi\rangle$ and $|\phi\rangle$, $\langle\psi|\phi\rangle = \langle\hat{U}\psi|\hat{U}\phi\rangle$.
 - norms, i.e., for any vector $|\psi\rangle$, $\| |\psi\rangle \| = \| \hat{U} |\psi\rangle \|$.
 - commutators, i.e., for any two operators $\hat{A}$ and $\hat{B}$, $[\hat{A}, \hat{B}] = [\hat{U}\hat{A}\hat{U}^\dagger, \hat{U}\hat{B}\hat{U}^\dagger]$.
 - eigenvalues, i.e., for any operator $\hat{A}$ with eigenvalues λ_i , the eigenvalues of $\hat{U}\hat{A}\hat{U}^\dagger$ are also λ_i .
 - traces, i.e., for any operator $\hat{A}$, $\text{Tr}(\hat{A}) = \text{Tr}(\hat{U}\hat{A}\hat{U}^\dagger)$.

Which of these hold for non-unitary similarity transformations? ($\hat{A} \mapsto \hat{A}' = \hat{S}\hat{A}\hat{S}^{-1}$ with $\hat{S}$ not necessarily unitary but invertible.)



Read S&N Sec. 1.7.2 - 1.7.5.



Question 2) Consider a more general operator $\hat{V} = e^{i\hat{G}}$ for some $\hat{G} = \hat{H} + i\hat{\Gamma}$, where both $\hat{H}$ and $\hat{\Gamma}$ are Hermitian. Assume that $[\hat{H}, \hat{\Gamma}] = 0$, so that they share a common eigenbasis.

- Under what condition is $\hat{V}$ unitary?
- From now on, consider the special case when $\hat{\Gamma} = \Gamma\hat{1}$ for some real constant Γ . Interpreting $\hat{H} + i\hat{\Gamma}$ as a complexified momentum operator (i.e., $\hat{p} \mapsto \hat{\mathcal{P}} = \hat{p} + i\hat{\Gamma}$), how does the translation operator

$$\hat{\mathcal{J}}(\Delta x) \rightarrow \lim_{N \rightarrow \infty} \left(\hat{1} - i \frac{\hat{\mathcal{P}} \Delta x}{N\hbar} \right)^N \quad (1)$$

act on a state $|\alpha\rangle$? What happens to the norm of the state under translations when $\hat{\Gamma} \neq 0$?

- Find the general expression for the “complexified” momentum operator $\hat{p} + i\hat{\Gamma}$ on a state α , generalizing S&N Eq. 1.249 and S&N Eq. 1.252. That is, find the generalization of

$$\langle x' | \hat{p}^n | \alpha \rangle = (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \langle x' | \alpha \rangle = (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \psi_\alpha(x') \quad \text{for } \hat{p} \mapsto \hat{\mathcal{P}} \quad (2)$$

- What is the expectation value of the momentum $\langle \hat{\mathcal{P}} \rangle_\alpha$ and kinetic energy of our state $\langle \hat{T} \rangle_\alpha = \langle \frac{\hat{\mathcal{P}}^2}{2m} \rangle_\alpha$ in terms of $\langle \hat{p} \rangle_\alpha$ and $\langle \hat{p}^2 \rangle_\alpha$? Can you reflect on what this would mean physically? (Hint: we will see later that for a free-particle the time evolution of a state is given by $|\alpha(t)\rangle = e^{-i\hat{T}t/\hbar} |\alpha\rangle$.)